Microeconomics

Lecture 2 · The consumer problem

Arseniy Stolyarov

PhD in Economics
University of Wisconsin–Madison

Artificial Intelligence and Financial Technologies

New Economic School · 2026/27

From preferences to an actual choice

- Budget sets, existence and Marshallian demand.

From preferences to an actual choice

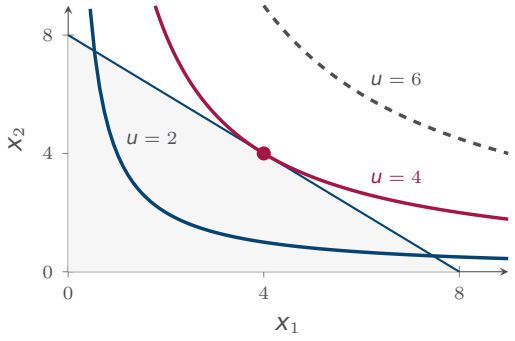
NES

- Budget sets, existence and Marshallian demand.
- The Lagrangian, saddle points and Kuhn–Tucker conditions, with two proofs.

From preferences to an actual choice

- Budget sets, existence and Marshallian demand.
- The Lagrangian, saddle points and Kuhn–Tucker conditions, with two proofs.
- Budget exhaustion and worked examples; then indirect utility, the envelope theorem and Roy’s identity.

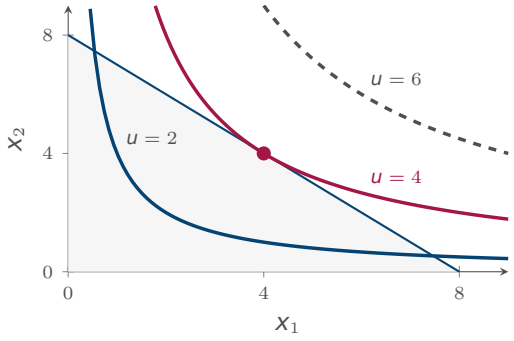
The picture we already know



$u(x) = \sqrt{x_1 x_2}$.

$p = (1, 1)$, $m = 8$.

The picture we already know

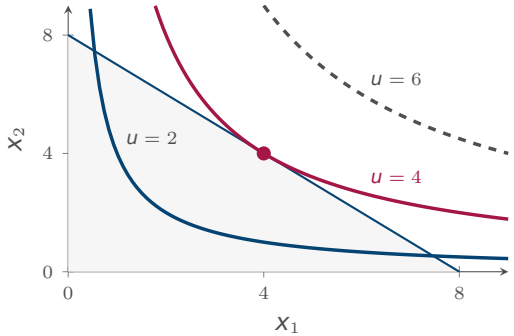


$u(x) = \sqrt{x_1 x_2}$.

$p = (1, 1)$, $m = 8$.

Which affordable bundle reaches the highest indifference curve?

The picture we already know



$u(x) = \sqrt{x_1 x_2}$.

$p = (1, 1)$, $m = 8$.

Which affordable bundle reaches the highest indifference curve?

Best bundle: $(4, 4)$.

Utility is 4.

The consumer problem

$$\max_{x} \quad u(x)$$

The consumer problem

$$\max_{x} \quad u(x)$$

$$\text{subject to} \quad p \cdot x \leq m,$$

The consumer problem

$$\begin{aligned} \max_{x} \quad & u(x) \\ \text{subject to} \quad & p \cdot x \leq m, \\ & x_i \geq 0 \quad \text{for every } i. \end{aligned}$$

The consumer problem

$$\begin{aligned} \max_{x} \quad & u(x) \\ \text{subject to} \quad & p \cdot x \leq m, \\ & x_i \geq 0 \quad \text{for every } i. \end{aligned}$$

Here $p \gg 0$ and $m > 0$. The feasible set is

$$B(p, m) = \{x \geq 0 : p \cdot x \leq m\}.$$

What did we assume when we wrote that?

- **Price-taking:** prices are given rather than negotiated.

What did we assume when we wrote that?

NES

- **Price-taking:** prices are given rather than negotiated.
- **Linear prices:** every unit has the same unit price.

What did we assume when we wrote that?

NES

- **Price-taking:** prices are given rather than negotiated.
- **Linear prices:** every unit has the same unit price.
- **Divisibility:** quantities can vary continuously.

What did we assume when we wrote that?

- **Price-taking:** prices are given rather than negotiated.
- **Linear prices:** every unit has the same unit price.
- **Divisibility:** quantities can vary continuously.
- **Decision horizon:** include saving or future consumption when relevant.

What did we assume when we wrote that?

- **Price-taking:** prices are given rather than negotiated.
- **Linear prices:** every unit has the same unit price.
- **Divisibility:** quantities can vary continuously.
- **Decision horizon:** include saving or future consumption when relevant.

What changes for a subscription, a quantity discount or an indivisible purchase?

Currency invariance of the budget

Claim. For every $t > 0$,

$$B(tp, tm) = B(p, m).$$

Currency invariance of the budget

Claim. For every $t > 0$,

$$B(tp, tm) = B(p, m).$$

Proof. For any $x \geq 0$,

Currency invariance of the budget

Claim. For every $t > 0$,

$$B(tp, tm) = B(p, m).$$

Proof. For any $x \geq 0$,

$$tp \cdot x \leq tm \quad \Longleftrightarrow \quad p \cdot x \leq m.$$

Currency invariance of the budget

Claim. For every $t > 0$,

$$B(tp, tm) = B(p, m).$$

Proof. For any $x \geq 0$,

$$tp \cdot x \leq tm \quad \Longleftrightarrow \quad p \cdot x \leq m.$$

The same bundles are affordable. Only the currency units have changed.

The budget is nonempty and compact

Claim. If $p \gg 0$ and $m > 0$, then $B(p, m)$ is nonempty and compact.

The budget is nonempty and compact

Claim. If $p \gg 0$ and $m > 0$, then $B(p, m)$ is nonempty and compact.

Proof: nonempty. The zero bundle is affordable.

The budget is nonempty and compact

Claim. If $p \gg 0$ and $m > 0$, then $B(p, m)$ is nonempty and compact.

Proof: nonempty. The zero bundle is affordable.

Closed. The inequalities $x_i \geq 0$ and $p \cdot x \leq m$ define closed sets.

The budget is nonempty and compact

Claim. If $p \gg 0$ and $m > 0$, then $B(p, m)$ is nonempty and compact.

Proof: nonempty. The zero bundle is affordable.

Closed. The inequalities $x_i \geq 0$ and $p \cdot x \leq m$ define closed sets.

Bounded. Every affordable bundle satisfies $0 \leq x_i \leq m/p_i$ for every i.

The budget is nonempty and compact

Claim. If $p \gg 0$ and $m > 0$, then $B(p, m)$ is nonempty and compact.

Proof: nonempty. The zero bundle is affordable.

Closed. The inequalities $x_i \geq 0$ and $p \cdot x \leq m$ define closed sets.

Bounded. Every affordable bundle satisfies $0 \leq x_i \leq m/p_i$ for every i.

A closed and bounded subset of finite-dimensional Euclidean space is compact.

Does an optimum exist?

Assumptions: u is continuous on $\mathbb{R}^L_+$, $p \gg 0$ and $m > 0$.

Does an optimum exist?

Assumptions: u is continuous on $\mathbb{R}^L_+$, $p \gg 0$ and $m > 0$.

The budget is nonempty and compact.

Does an optimum exist?

Assumptions: u is continuous on $\mathbb{R}^L_+$, $p \gg 0$ and $m > 0$.

The budget is nonempty and compact.

By the extreme value theorem, a continuous function attains a maximum on it.

Does an optimum exist?

Assumptions: u is continuous on $\mathbb{R}^L_+$, $p \gg 0$ and $m > 0$.

The budget is nonempty and compact.

By the extreme value theorem, a continuous function attains a maximum on it.

Existence does not yet imply uniqueness, an interior solution or differentiability of demand.

Which assumption is doing the work?

Suppose $p_1 = 0$, $p_2 = 1$, $m = 1$, and $u(x) = x_1$.

Which assumption is doing the work?

Suppose $p_1 = 0$, $p_2 = 1$, $m = 1$, and $u(x) = x_1$.

Every quantity of good 1 is affordable. Utility is unbounded above, so there is no maximiser.

Which assumption is doing the work?

Suppose $p_1 = 0$, $p_2 = 1$, $m = 1$, and $u(x) = x_1$.

Every quantity of good 1 is affordable. Utility is unbounded above, so there is no maximiser.

Zero prices do not always destroy existence, but compactness is no longer guaranteed.

Which assumption is doing the work?

Suppose $p_1 = 0$, $p_2 = 1$, $m = 1$, and $u(x) = x_1$.

Every quantity of good 1 is affordable. Utility is unbounded above, so there is no maximiser.

Zero prices do not always destroy existence, but compactness is no longer guaranteed.

Can you give a case with multiple maximisers even though the budget is compact?

Which assumption is doing the work?

Suppose $p_1 = 0$, $p_2 = 1$, $m = 1$, and $u(x) = x_1$.

Every quantity of good 1 is affordable. Utility is unbounded above, so there is no maximiser.

Zero prices do not always destroy existence, but compactness is no longer guaranteed.

Can you give a case with multiple maximisers even though the budget is compact?

For $u = x_1 + x_2$, $p = (1, 1)$ and $m = 1$, every bundle on $x_1 + x_2 = 1$ is optimal.

Marshallian demand is initially a correspondence

$$D(p, m) = \arg\max_{x \geq 0,\, p \cdot x \leq m} u(x).$$

Marshallian demand is initially a correspondence

$$D(p, m) = \underset{x \geq 0,\, p \cdot x \leq m}{\arg\max}\, u(x).$$

$D(p, m)$ contains **all** optimal bundles. It may contain more than one.

Marshallian demand is initially a correspondence

NES

$$D(p, m) = \underset{x \geq 0,\, p \cdot x \leq m}{\arg\max}\, u(x).$$

$D(p, m)$ contains **all** optimal bundles. It may contain more than one.

When it is a singleton, write its unique element as $x(p, m)$: a demand function.

Marshallian demand is initially a correspondence

$$D(p, m) = \underset{x \geq 0,\, p \cdot x \leq m}{\arg\max}\, u(x).$$

$D(p, m)$ contains **all** optimal bundles. It may contain more than one.

When it is a singleton, write its unique element as $x(p, m)$: a demand function.

It is the earlier choice rule applied specifically to budget menus: $D(p, m) = C(B(p, m))$.

Proof: demand is homogeneous of degree zero

Claim: $D(tp, tm) = D(p, m)$ for every $t > 0$.

Proof: demand is homogeneous of degree zero

Claim: $D(tp, tm) = D(p, m)$ for every $t > 0$.

$tp \cdot x \leq tm \iff p \cdot x \leq m$: divide by the positive number t.

Proof: demand is homogeneous of degree zero

Claim: $D(tp, tm) = D(p, m)$ for every $t > 0$.

$tp \cdot x \leq tm \iff p \cdot x \leq m$: divide by the positive number t.

Therefore $B(tp, tm) = B(p, m)$. The objective u is the same.

Proof: demand is homogeneous of degree zero

Claim: $D(tp, tm) = D(p, m)$ for every $t > 0$.

$tp \cdot x \leq tm \iff p \cdot x \leq m$: divide by the positive number t.

Therefore $B(tp, tm) = B(p, m)$. The objective u is the same.

The full argmax sets coincide. Neither uniqueness nor differentiability is needed.

Proof: demand is homogeneous of degree zero

Claim: $D(tp, tm) = D(p, m)$ for every $t > 0$.

$tp \cdot x \le tm \iff p \cdot x \le m$: divide by the positive number t.

Therefore $B(tp, tm) = B(p, m)$. The objective u is the same.

The full argmax sets coincide. Neither uniqueness nor differentiability is needed.

Example: $(p, m) = ((2, 1), 12)$ and $((200, 100), 1200)$.

Why introduce a Lagrangian?

NES

Write every inequality as a nonnegative slack: $m - p \cdot x \geq 0$ and $x_i \geq 0$.

Why introduce a Lagrangian?

Write every inequality as a nonnegative slack: $m - p \cdot x \geq 0$ and $x_i \geq 0$.

$$\mathcal{L}(x, \lambda, \mu) = u(x) + \lambda(m - p \cdot x) + \mu \cdot x, \qquad \lambda, \mu \geq 0.$$

Why introduce a Lagrangian?

Write every inequality as a nonnegative slack: $m - p \cdot x \geq 0$ and $x_i \geq 0$.

$$\mathcal{L}(x, \lambda, \mu) = u(x) + \lambda(m - p \cdot x) + \mu \cdot x, \qquad \lambda, \mu \geq 0.$$

A violated constraint lets its multiplier drive $\mathcal{L}$ down. At a feasible bundle the added terms are nonnegative.

Why introduce a Lagrangian?

Write every inequality as a nonnegative slack: $m - p \cdot x \geq 0$ and $x_i \geq 0$.

$$\mathcal{L}(x, \lambda, \mu) = u(x) + \lambda(m - p \cdot x) + \mu \cdot x, \qquad \lambda, \mu \geq 0.$$

A violated constraint lets its multiplier drive $\mathcal{L}$ down. At a feasible bundle the added terms are nonnegative.

For the saddle argument, vary x in an open convex domain U where u is defined.

What is a saddle point?

A triple (x^*, λ^*, μ^*) is a global saddle if, for every $x \in U$ and $\lambda, \mu \geq 0$,

What is a saddle point?

A triple (x^*, λ^*, μ^*) is a global saddle if, for every $x \in U$ and $\lambda, \mu \geq 0$,

$$\mathcal{L}(x, \lambda^*, \mu^*) \leq \mathcal{L}(x^*, \lambda^*, \mu^*) \leq \mathcal{L}(x^*, \lambda, \mu).$$

What is a saddle point?

A triple (x^*, λ^*, μ^*) is a global saddle if, for every $x \in U$ and $\lambda, \mu \geq 0$,

$$\mathcal{L}(x, \lambda^*, \mu^*) \leq \mathcal{L}(x^*, \lambda^*, \mu^*) \leq \mathcal{L}(x^*, \lambda, \mu).$$

With multipliers fixed, x^* maximises the Lagrangian.

What is a saddle point?

A triple (x^*, λ^*, μ^*) is a global saddle if, for every $x \in U$ and $\lambda, \mu \geq 0$,

$$\mathcal{L}(x, \lambda^*, \mu^*) \leq \mathcal{L}(x^*, \lambda^*, \mu^*) \leq \mathcal{L}(x^*, \lambda, \mu).$$

With multipliers fixed, x^* maximises the Lagrangian.

With quantities fixed, (λ^*, μ^*) minimises it over nonnegative multipliers.

Theorem 1: a saddle gives a consumer optimum

If (x^*, λ^*, μ^*) is a global saddle, then x^* maximises u over $x \in U$, $x \geq 0$, $p \cdot x \leq m$. We show:

Theorem 1: a saddle gives a consumer optimum

If (x^*, λ^*, μ^*) is a global saddle, then x^* maximises u over $x \in U$, $x \geq 0$, $p \cdot x \leq m$. We show:

1. **Feasibility:** $x^* \geq 0$, $p \cdot x^* \leq m$.

Theorem 1: a saddle gives a consumer optimum

If (x^*, λ^*, μ^*) is a global saddle, then x^* maximises u over $x \in U$, $x \geq 0$, $p \cdot x \leq m$. We show:

1. **Feasibility:** $x^* \geq 0$, $p \cdot x^* \leq m$.
2. **Slackness:** $\lambda^*(m - p \cdot x^*) = 0$, $\mu_i^* x_i^* = 0$.

Theorem 1: a saddle gives a consumer optimum

If (x^*, λ^*, μ^*) is a global saddle, then x^* maximises u over $x \in U$, $x \geq 0$, $p \cdot x \leq m$. We show:

1. **Feasibility:** $x^* \geq 0$, $p \cdot x^* \leq m$.
2. **Slackness:** $\lambda^*(m - p \cdot x^*) = 0$, $\mu_i^* x_i^* = 0$.
3. **Value:** $\mathcal{L}(x^*, \lambda^*, \mu^*) = u(x^*)$.

Theorem 1: a saddle gives a consumer optimum

If (x^*, λ^*, μ^*) is a global saddle, then x^* maximises u over $x \in U$, $x \geq 0$, $p \cdot x \leq m$. We show:

1. **Feasibility:** $x^* \geq 0$, $p \cdot x^* \leq m$.
2. **Slackness:** $\lambda^*(m - p \cdot x^*) = 0$, $\mu_i^* x_i^* = 0$.
3. **Value:** $\mathcal{L}(x^*, \lambda^*, \mu^*) = u(x^*)$.
4. **Optimality:** every feasible y satisfies $u(y) \leq u(x^*)$.

Proof 1: feasibility

At the saddle, nonnegative multipliers minimise $\mathcal{L}(x^*, \lambda, \mu)$.

Proof 1: feasibility

At the saddle, nonnegative multipliers minimise $\mathcal{L}(x^*, \lambda, \mu)$.

If $p \cdot x^* > m$, increasing λ sends $\mathcal{L}$ to $-\infty$: impossible.

Proof 1: feasibility

At the saddle, nonnegative multipliers minimise $\mathcal{L}(x^*, \lambda, \mu)$.

If $p \cdot x^* > m$, increasing λ sends $\mathcal{L}$ to $-\infty$: impossible.

If some $x_i^* < 0$, increasing μ_i sends $\mathcal{L}$ to $-\infty$: also impossible.

Proof 1: feasibility

At the saddle, nonnegative multipliers minimise $\mathcal{L}(x^*, \lambda, \mu)$.

If $p \cdot x^* > m$, increasing λ sends $\mathcal{L}$ to $-\infty$: impossible.

If some $x_i^* < 0$, increasing μ_i sends $\mathcal{L}$ to $-\infty$: also impossible.

Therefore $p \cdot x^* \leq m$ and $x^* \geq 0$.

Proof 1: slackness and equality of values

If $\lambda^* > 0$ and $m - p \cdot x^* > 0$, reducing λ to zero lowers $\mathcal{L}$: a contradiction.

Proof 1: slackness and equality of values

If $\lambda^* > 0$ and $m - p \cdot x^* > 0$, reducing λ to zero lowers $\mathcal{L}$: a contradiction.

Likewise $\mu_i^* > 0$ and $x_i^* > 0$ cannot hold together.

Proof 1: slackness and equality of values

If $\lambda^* > 0$ and $m - p \cdot x^* > 0$, reducing λ to zero lowers $\mathcal{L}$: a contradiction.

Likewise $\mu_i^* > 0$ and $x_i^* > 0$ cannot hold together.

Thus every multiplier–slack product is zero:

Proof 1: slackness and equality of values

If $\lambda^* > 0$ and $m - p \cdot x^* > 0$, reducing λ to zero lowers $\mathcal{L}$: a contradiction.

Likewise $\mu_i^* > 0$ and $x_i^* > 0$ cannot hold together.

Thus every multiplier–slack product is zero:

$$\mathcal{L}(x^*, \lambda^*, \mu^*) = u(x^*) + \underbrace{\lambda^*(m - p \cdot x^*)}_{0} + \underbrace{\mu^* \cdot x^*}_{0} = u(x^*).$$

Proof 1: compare with any affordable bundle

For every feasible $y \in U$, $\lambda^*, \mu^* \geq 0$ give $u(y) \leq \mathcal{L}(y, \lambda^*, \mu^*)$.

Proof 1: compare with any affordable bundle

NES

For every feasible $y \in U$, $\lambda^*, \mu^* \geq 0$ give $u(y) \leq \mathcal{L}(y, \lambda^*, \mu^*)$.

The saddle inequality then gives

Proof 1: compare with any affordable bundle

For every feasible $y \in U$, $\lambda^*, \mu^* \geq 0$ give $u(y) \leq \mathcal{L}(y, \lambda^*, \mu^*)$.

The saddle inequality then gives

$$u(y) \leq \mathcal{L}(y, \lambda^*, \mu^*) \leq \mathcal{L}(x^*, \lambda^*, \mu^*) = u(x^*).$$

Proof 1: compare with any affordable bundle

NES

For every feasible $y \in U$, $\lambda^*, \mu^* \geq 0$ give $u(y) \leq \mathcal{L}(y, \lambda^*, \mu^*)$.

The saddle inequality then gives

$$u(y) \leq \mathcal{L}(y, \lambda^*, \mu^*) \leq \mathcal{L}(x^*, \lambda^*, \mu^*) = u(x^*).$$

This proves optimality. This direction needs no concavity or differentiability.

Theorem 2: an optimum yields a saddle

Let $p \gg 0$, $m > 0$, and let u be differentiable and concave on U. For every consumer optimum x^*, suitable nonnegative multipliers give a global saddle. We show:

Theorem 2: an optimum yields a saddle

Let $p \gg 0$, $m > 0$, and let u be differentiable and concave on U. For every consumer optimum x^*, suitable nonnegative multipliers give a global saddle. We show:

1. A common, nonnegative budget multiplier λ^*.

Theorem 2: an optimum yields a saddle

Let $p \gg 0$, $m > 0$, and let u be differentiable and concave on U. For every consumer optimum x^*, suitable nonnegative multipliers give a global saddle. We show:

1. A common, nonnegative budget multiplier λ^*.
2. Nonnegative quantity-bound multipliers μ_i^*.

Theorem 2: an optimum yields a saddle

Let $p \gg 0$, $m > 0$, and let u be differentiable and concave on U. For every consumer optimum x^*, suitable nonnegative multipliers give a global saddle. We show:

1. A common, nonnegative budget multiplier λ^*.
2. Nonnegative quantity-bound multipliers μ_i^*.
3. Stationarity and complementary slackness.

Theorem 2: an optimum yields a saddle

Let $p \gg 0$, $m > 0$, and let u be differentiable and concave on U. For every consumer optimum x^*, suitable nonnegative multipliers give a global saddle. We show:

1. A common, nonnegative budget multiplier λ^*.
2. Nonnegative quantity-bound multipliers μ_i^*.
3. Stationarity and complementary slackness.
4. Maximisation in x and minimisation in the multipliers.

Proof 2: construct the multipliers

First take $x^* \neq 0$. Choose a consumed good j, with $x_j^* > 0$.

Proof 2: construct the multipliers

First take $x^* \neq 0$. Choose a consumed good j, with $x_j^* > 0$.

$$\lambda^* := \frac{u_j(x^*)}{p_j}, \qquad \mu_i^* := \begin{cases} 0, & x_i^* > 0, \\ \lambda^* p_i - u_i(x^*), & x_i^* = 0. \end{cases}$$

Proof 2: construct the multipliers

First take $x^* \neq 0$. Choose a consumed good j, with $x_j^* > 0$.

$$\lambda^* := \frac{u_j(x^*)}{p_j}, \qquad \mu_i^* := \begin{cases} 0, & x_i^* > 0, \\ \lambda^* p_i - u_i(x^*), & x_i^* = 0. \end{cases}$$

Here $u_i = \partial u / \partial x_i$. The zero-bundle case will be handled separately.

Proof 2: a common nonnegative spending return

If $\lambda^* < 0$, then $u_j(x^*) < 0$: slightly reducing consumed good j improves utility and remains affordable. Hence $\lambda^* \geq 0$.

Proof 2: a common nonnegative spending return

If $\lambda^* < 0$, then $u_j(x^*) < 0$: slightly reducing consumed good j improves utility and remains affordable. Hence $\lambda^* \geq 0$.

If two consumed goods had different u_i/p_i, transfer a small amount of spending toward the larger return.

Proof 2: a common nonnegative spending return

If $\lambda^* < 0$, then $u_j(x^*) < 0$: slightly reducing consumed good j improves utility and remains affordable. Hence $\lambda^* \geq 0$.

If two consumed goods had different u_i/p_i, transfer a small amount of spending toward the larger return.

The transfer is feasible in either direction and has positive utility derivative, contradicting optimality.

Proof 2: a common nonnegative spending return

If $\lambda^* < 0$, then $u_j(x^*) < 0$: slightly reducing consumed good j improves utility and remains affordable. Hence $\lambda^* \geq 0$.

If two consumed goods had different u_i/p_i, transfer a small amount of spending toward the larger return.

The transfer is feasible in either direction and has positive utility derivative, contradicting optimality.

Therefore $u_i(x^*)/p_i = \lambda^*$ for every consumed good, independently of which j was selected.

Proof 2: why a negative bound multiplier fails

Only $x_i^* = 0$ needs checking: for a consumed good we already set $\mu_i^* = 0$.

Proof 2: why a negative bound multiplier fails

Only $x_i^* = 0$ needs checking: for a consumed good we already set $\mu_i^* = 0$.

Suppose $\mu_i^* < 0$. By its definition,

$$\lambda^* p_i - u_i(x^*) < 0 \quad \Rightarrow \quad \lambda^* < \frac{u_i(x^*)}{p_i}.$$

Proof 2: why a negative bound multiplier fails

Only $x_i^* = 0$ needs checking: for a consumed good we already set $\mu_i^* = 0$.

Suppose $\mu_i^* < 0$. By its definition,

$$\lambda^* p_i - u_i(x^*) < 0 \quad \Rightarrow \quad \lambda^* < \frac{u_i(x^*)}{p_i}.$$

For the consumed good j, $\lambda^* = u_j(x^*)/p_j$. Therefore

$$\frac{u_i(x^*)}{p_i} > \frac{u_j(x^*)}{p_j}.$$

Proof 2: why a negative bound multiplier fails

Only $x_i^* = 0$ needs checking: for a consumed good we already set $\mu_i^* = 0$.

Suppose $\mu_i^* < 0$. By its definition,

$$\lambda^* p_i - u_i(x^*) < 0 \quad \Rightarrow \quad \lambda^* < \frac{u_i(x^*)}{p_i}.$$

For the consumed good j, $\lambda^* = u_j(x^*)/p_j$. Therefore

$$\frac{u_i(x^*)}{p_i} > \frac{u_j(x^*)}{p_j}.$$

The unconsumed good would give more marginal utility per unit of spending.

Proof 2: the profitable spending transfer

Shift spending ε from consumed good j to unconsumed good i:

Proof 2: the profitable spending transfer

Shift spending ε from consumed good j to unconsumed good i:

$$x_i(\varepsilon) = \frac{\varepsilon}{p_i}, \qquad x_j(\varepsilon) = x_j^* - \frac{\varepsilon}{p_j}.$$

Proof 2: the profitable spending transfer

NES

Shift spending ε from consumed good j to unconsumed good i:

$$x_i(\varepsilon) = \frac{\varepsilon}{p_i}, \qquad x_j(\varepsilon) = x_j^* - \frac{\varepsilon}{p_j}.$$

Keep other quantities fixed. For $0 < \varepsilon < p_j x_j^*$, quantities remain nonnegative and total spending is unchanged.

Proof 2: the profitable spending transfer

NES

Shift spending ε from consumed good j to unconsumed good i:

$$x_i(\varepsilon) = \frac{\varepsilon}{p_i}, \qquad x_j(\varepsilon) = x_j^* - \frac{\varepsilon}{p_j}.$$

Keep other quantities fixed. For $0 < \varepsilon < p_j x_j^*$, quantities remain nonnegative and total spending is unchanged.

$$\frac{d}{d\varepsilon} u(x(\varepsilon))\bigg|_{0+} = \frac{u_i(x^*)}{p_i} - \frac{u_j(x^*)}{p_j} = -\frac{\mu_i^*}{p_i} > 0.$$

Proof 2: the profitable spending transfer

Shift spending ε from consumed good j to unconsumed good i:

$$x_i(\varepsilon) = \frac{\varepsilon}{p_i}, \qquad x_j(\varepsilon) = x_j^* - \frac{\varepsilon}{p_j}.$$

Keep other quantities fixed. For $0 < \varepsilon < p_j x_j^*$, quantities remain nonnegative and total spending is unchanged.

$$\frac{d}{d\varepsilon} u(x(\varepsilon))\bigg|_{0+} = \frac{u_i(x^*)}{p_i} - \frac{u_j(x^*)}{p_j} = -\frac{\mu_i^*}{p_i} > 0.$$

A sufficiently small transfer strictly improves utility: contradiction. Thus $\mu_i^* \geq 0$, with equality allowed.

Proof 2: stationarity and complementary slackness

If $p \cdot x^* = m$, then $\lambda^*(m - p \cdot x^*) = 0$.

Proof 2: stationarity and complementary slackness

If $p \cdot x^* = m$, then $\lambda^*(m - p \cdot x^*) = 0$.

If the budget is slack and $x_j^* > 0$, both a small increase and a small decrease in x_j are feasible.

Proof 2: stationarity and complementary slackness

If $p \cdot x^* = m$, then $\lambda^*(m - p \cdot x^*) = 0$.

If the budget is slack and $x_j^* > 0$, both a small increase and a small decrease in x_j are feasible.

Local optimality forces $u_j(x^*) = 0$, hence $\lambda^* = u_j(x^*)/p_j = 0$. The budget product is again zero.

Proof 2: stationarity and complementary slackness

If $p \cdot x^* = m$, then $\lambda^*(m - p \cdot x^*) = 0$.

If the budget is slack and $x_j^* > 0$, both a small increase and a small decrease in x_j are feasible.

Local optimality forces $u_j(x^*) = 0$, hence $\lambda^* = u_j(x^*)/p_j = 0$. The budget product is again zero.

For every i, the construction gives $\mu_i^* x_i^* = 0$: either the quantity or its multiplier is zero.

Proof 2: stationarity and complementary slackness

If $p \cdot x^* = m$, then $\lambda^*(m - p \cdot x^*) = 0$.

If the budget is slack and $x_j^* > 0$, both a small increase and a small decrease in x_j are feasible.

Local optimality forces $u_j(x^*) = 0$, hence $\lambda^* = u_j(x^*)/p_j = 0$. The budget product is again zero.

For every i, the construction gives $\mu_i^* x_i^* = 0$: either the quantity or its multiplier is zero.

The construction also gives $u_i(x^*) - \lambda^* p_i + \mu_i^* = 0$.

Proof 2: maximise in quantities

Fix the constructed multipliers and put $f(x) = \mathcal{L}(x, \lambda^*, \mu^*)$.

Proof 2: maximise in quantities

Fix the constructed multipliers and put $f(x) = \mathcal{L}(x, \lambda^*, \mu^*)$.

Concave u plus affine terms gives concave f on U. Stationarity gives $\nabla f(x^*) = 0$.

Proof 2: maximise in quantities

Fix the constructed multipliers and put $f(x) = \mathcal{L}(x, \lambda^*, \mu^*)$.

Concave u plus affine terms gives concave f on U. Stationarity gives $\nabla f(x^*) = 0$.

A differentiable concave function lies below its tangent plane:

Proof 2: maximise in quantities

Fix the constructed multipliers and put $f(x) = \mathcal{L}(x, \lambda^*, \mu^*)$.

Concave u plus affine terms gives concave f on U. Stationarity gives $\nabla f(x^*) = 0$.

A differentiable concave function lies below its tangent plane:

$$f(y) \leq f(x^*) + \nabla f(x^*) \cdot (y - x^*) = f(x^*) \qquad (y \in U).$$

Proof 2: if the optimal bundle is zero

If $x^* = 0 \in U$, a small increase in any good is affordable since $p \gg 0$, $m > 0$ and U is open.

Proof 2: if the optimal bundle is zero

If $x^* = 0 \in U$, a small increase in any good is affordable since $p \gg 0$, $m > 0$ and U is open.

Optimality implies $u_i(0) \leq 0$ for every i.

Proof 2: if the optimal bundle is zero

If $x^* = 0 \in U$, a small increase in any good is affordable since $p \gg 0$, $m > 0$ and U is open.

Optimality implies $u_i(0) \leq 0$ for every i.

Choose $\lambda^* = 0$, $\mu_i^* = -u_i(0) \geq 0$. Stationarity and complementary slackness hold.

Proof 2: if the optimal bundle is zero

If $x^* = 0 \in U$, a small increase in any good is affordable since $p \gg 0$, $m > 0$ and U is open.

Optimality implies $u_i(0) \leq 0$ for every i.

Choose $\lambda^* = 0$, $\mu_i^* = -u_i(0) \geq 0$. Stationarity and complementary slackness hold.

The same concavity argument makes $x^* = 0$ maximise the fixed-multiplier Lagrangian.

Proof 2: minimise in multipliers and finish

Feasibility and complementary slackness give $\mathcal{L}(x^*, \lambda^*, \mu^*) = u(x^*)$.

Proof 2: minimise in multipliers and finish

Feasibility and complementary slackness give $\mathcal{L}(x^*, \lambda^*, \mu^*) = u(x^*)$.

For any $\lambda, \mu \geq 0$, the nonnegative slacks give $\mathcal{L}(x^*, \lambda, \mu) \geq u(x^*)$.

Proof 2: minimise in multipliers and finish

Feasibility and complementary slackness give $\mathcal{L}(x^*, \lambda^*, \mu^*) = u(x^*)$.

For any $\lambda, \mu \geq 0$, the nonnegative slacks give $\mathcal{L}(x^*, \lambda, \mu) \geq u(x^*)$.

Combining this with maximisation in quantities,

Proof 2: minimise in multipliers and finish

Feasibility and complementary slackness give $\mathcal{L}(x^*, \lambda^*, \mu^*) = u(x^*)$.

For any $\lambda, \mu \geq 0$, the nonnegative slacks give $\mathcal{L}(x^*, \lambda, \mu) \geq u(x^*)$.

Combining this with maximisation in quantities,

$$\mathcal{L}(x, \lambda^*, \mu^*) \leq \mathcal{L}(x^*, \lambda^*, \mu^*) \leq \mathcal{L}(x^*, \lambda, \mu).$$

Proof 2: minimise in multipliers and finish

Feasibility and complementary slackness give $\mathcal{L}(x^*, \lambda^*, \mu^*) = u(x^*)$.

For any $\lambda, \mu \geq 0$, the nonnegative slacks give $\mathcal{L}(x^*, \lambda, \mu) \geq u(x^*)$.

Combining this with maximisation in quantities,

$$\mathcal{L}(x, \lambda^*, \mu^*) \leq \mathcal{L}(x^*, \lambda^*, \mu^*) \leq \mathcal{L}(x^*, \lambda, \mu).$$

The constructed triple is a saddle. Both nonzero and zero optima are covered.

The four Kuhn–Tucker groups

Stationarity: $u_i(x^*) - \lambda^* p_i + \mu_i^* = 0.$

The four Kuhn–Tucker groups

NES

Stationarity: $u_i(x^*) - \lambda^* p_i + \mu_i^* = 0.$

Primal feasibility: $p \cdot x^* \leq m,\ x^* \geq 0.$

The four Kuhn–Tucker groups

Stationarity: $u_i(x^*) - \lambda^* p_i + \mu_i^* = 0.$

Primal feasibility: $p \cdot x^* \leq m$, $x^* \geq 0.$

Dual feasibility: $\lambda^* \geq 0$, $\mu^* \geq 0.$

The four Kuhn–Tucker groups

Stationarity: $u_i(x^*) - \lambda^* p_i + \mu_i^* = 0.$

Primal feasibility: $p \cdot x^* \leq m$, $x^* \geq 0.$

Dual feasibility: $\lambda^* \geq 0$, $\mu^* \geq 0.$

Complementary slackness:

$$\lambda^*(m - p \cdot x^*) = 0, \qquad \mu_i^* x_i^* = 0.$$

Example 1: Cobb–Douglas

$$u(x) = x_1^{\alpha} x_2^{1-\alpha}, \qquad 0 < \alpha < 1.$$

Example 1: Cobb–Douglas

$$u(x) = x_1^{\alpha} x_2^{1-\alpha}, \qquad 0 < \alpha < 1.$$

Could an optimum lie on an axis? Which representation would make the calculation easier?

Example 1: Cobb–Douglas

$$u(x) = x_1^{\alpha} x_2^{1-\alpha}, \qquad 0 < \alpha < 1.$$

Could an optimum lie on an axis? Which representation would make the calculation easier?

An axis bundle gives zero utility; a strictly positive affordable bundle gives positive utility.

Example 1: Cobb–Douglas

$$u(x) = x_1^{\alpha} x_2^{1-\alpha}, \qquad 0 < \alpha < 1.$$

Could an optimum lie on an axis? Which representation would make the calculation easier?

An axis bundle gives zero utility; a strictly positive affordable bundle gives positive utility.

Every optimum is therefore positive. We may now take logs and use an interior method.

Choose a convenient representation

Maximising u is equivalent, at the positive optimum, to maximising

$$\widetilde{u}(x) = \alpha \log x_1 + (1 - \alpha) \log x_2.$$

Choose a convenient representation

Maximising u is equivalent, at the positive optimum, to maximising

$$\widetilde{u}(x) = \alpha \log x_1 + (1-\alpha) \log x_2.$$

$$\mathcal{L} = \alpha \log x_1 + (1-\alpha) \log x_2 + \lambda(m - p_1 x_1 - p_2 x_2).$$

Choose a convenient representation

Maximising u is equivalent, at the positive optimum, to maximising

$$\widetilde{u}(x) = \alpha \log x_1 + (1-\alpha) \log x_2.$$

$$\mathcal{L} = \alpha \log x_1 + (1-\alpha) \log x_2 + \lambda(m - p_1 x_1 - p_2 x_2).$$

The log representation is concave on the positive domain.

Choose a convenient representation

Maximising u is equivalent, at the positive optimum, to maximising

$$\widetilde{u}(x) = \alpha \log x_1 + (1-\alpha) \log x_2.$$

$$\mathcal{L} = \alpha \log x_1 + (1-\alpha) \log x_2 + \lambda(m - p_1 x_1 - p_2 x_2).$$

The log representation is concave on the positive domain.

The bundle does not change, but the numerical utility and multiplier generally do.

Solve the Cobb–Douglas conditions

$$\frac{\alpha}{x_1} = \lambda p_1, \qquad \frac{1-\alpha}{x_2} = \lambda p_2.$$

Solve the Cobb–Douglas conditions

$$\frac{\alpha}{x_1} = \lambda p_1, \qquad \frac{1-\alpha}{x_2} = \lambda p_2.$$

These imply $\lambda > 0$, so slackness forces $p \cdot x = m$.

Solve the Cobb–Douglas conditions

$$\frac{\alpha}{x_1} = \lambda p_1, \qquad \frac{1-\alpha}{x_2} = \lambda p_2.$$

These imply $\lambda > 0$, so slackness forces $p \cdot x = m$.

Adding $p_1 x_1 = \alpha/\lambda$ and $p_2 x_2 = (1-\alpha)/\lambda$ gives $\lambda = 1/m$.

Solve the Cobb–Douglas conditions

$$\frac{\alpha}{x_1} = \lambda p_1, \qquad \frac{1-\alpha}{x_2} = \lambda p_2.$$

These imply $\lambda > 0$, so slackness forces $p \cdot x = m$.

Adding $p_1 x_1 = \alpha/\lambda$ and $p_2 x_2 = (1-\alpha)/\lambda$ gives $\lambda = 1/m$.

$$x_1 = \frac{\alpha m}{p_1}, \qquad x_2 = \frac{(1-\alpha)m}{p_2}.$$

Solve the Cobb–Douglas conditions

$$\frac{\alpha}{x_1} = \lambda p_1, \qquad \frac{1-\alpha}{x_2} = \lambda p_2.$$

These imply $\lambda > 0$, so slackness forces $p \cdot x = m$.

Adding $p_1 x_1 = \alpha/\lambda$ and $p_2 x_2 = (1-\alpha)/\lambda$ gives $\lambda = 1/m$.

$$x_1 = \frac{\alpha m}{p_1}, \qquad x_2 = \frac{(1-\alpha)m}{p_2}.$$

At $\alpha = 1/2$, $p = (2, 1)$ and $m = 12$, demand is $(3, 6)$.

Walras' law: an optimum exhausts the budget

Claim. Under local nonsatiation, every existing optimum satisfies $p \cdot x^* = m$.

Walras' law: an optimum exhausts the budget

Claim. Under local nonsatiation, every existing optimum satisfies $p \cdot x^* = m$.

Proof. Suppose slack is $\delta = m - p \cdot x^* > 0$.

Walras' law: an optimum exhausts the budget

Claim. Under local nonsatiation, every existing optimum satisfies $p \cdot x^* = m$.

Proof. Suppose slack is $\delta = m - p \cdot x^* > 0$.

If $y \geq 0$ is within distance $\delta/(2\|p\|)$ of x^*, then $p \cdot y < m$.

Walras' law: an optimum exhausts the budget

Claim. Under local nonsatiation, every existing optimum satisfies $p \cdot x^* = m$.

Proof. Suppose slack is $\delta = m - p \cdot x^* > 0$.

If $y \geq 0$ is within distance $\delta/(2\|p\|)$ of x^*, then $p \cdot y < m$.

LNS supplies a strictly better such y, contradicting optimality.

Example 2: perfect substitutes

$$u(x) = ax_1 + bx_2, \qquad a, b > 0.$$

Example 2: perfect substitutes

$$u(x) = ax_1 + bx_2, \qquad a, b > 0.$$

Where would you spend the next unit of money? Can the optimum be nonunique?

Example 2: perfect substitutes

NES

$$u(x) = ax_1 + bx_2, \qquad a, b > 0.$$

Where would you spend the next unit of money? Can the optimum be nonunique?

Compare a/p_1 and b/p_2. Buy $(m/p_1, 0)$ if the first is larger; buy $(0, m/p_2)$ if the second is larger.

Example 2: perfect substitutes

$$u(x) = ax_1 + bx_2, \qquad a, b > 0.$$

Where would you spend the next unit of money? Can the optimum be nonunique?

Compare a/p_1 and b/p_2. Buy $(m/p_1, 0)$ if the first is larger; buy $(0, m/p_2)$ if the second is larger.

If the ratios are equal, every budget-line bundle is optimal.

Verify a corner with the full KKT system

Let $u(x) = 2x_1 + x_2$, $p = (3, 1)$ and $m = 12$.

Verify a corner with the full KKT system

Let $u(x) = 2x_1 + x_2$, $p = (3, 1)$ and $m = 12$.

The spending returns are $2/3$ and 1, so $x^* = (0, 12)$.

Verify a corner with the full KKT system

Let $u(x) = 2x_1 + x_2$, $p = (3, 1)$ and $m = 12$.

The spending returns are $2/3$ and 1, so $x^* = (0, 12)$.

Stationarity and slackness give $\lambda = 1$, $\mu_1 = 1$, $\mu_2 = 0$.

Verify a corner with the full KKT system

Let $u(x) = 2x_1 + x_2$, $p = (3, 1)$ and $m = 12$.

The spending returns are $2/3$ and 1, so $x^* = (0, 12)$.

Stationarity and slackness give $\lambda = 1$, $\mu_1 = 1$, $\mu_2 = 0$.

Check: $2 - 3\lambda + \mu_1 = 0$, $1 - \lambda + \mu_2 = 0$.

Verify a corner with the full KKT system

Let $u(x) = 2x_1 + x_2$, $p = (3, 1)$ and $m = 12$.

The spending returns are $2/3$ and 1, so $x^* = (0, 12)$.

Stationarity and slackness give $\lambda = 1$, $\mu_1 = 1$, $\mu_2 = 0$.

Check: $2 - 3\lambda + \mu_1 = 0$, $1 - \lambda + \mu_2 = 0$.

The zero quantity is optimal precisely because its marginal return is too low.

Example 3: lessons and money left over

$$u(a,q) = a + 4\log(1+q), \qquad a + rq \le m, \quad a, q \ge 0.$$

Example 3: lessons and money left over

NES

$$u(a,q) = a + 4\log(1+q), \qquad a + rq \le m, \quad a, q \ge 0.$$

The price of a is 1; the lesson price is $r > 0$; wealth is $m > 0$.

Example 3: lessons and money left over

$$u(a,q) = a + 4\log(1+q), \qquad a + rq \le m, \quad a, q \ge 0.$$

The price of a is 1; the lesson price is $r > 0$; wealth is $m > 0$.

Must both goods be consumed? How could we reduce this to one variable?

Example 3: lessons and money left over

NES

$$u(a,q) = a + 4\log(1+q), \qquad a + rq \le m, \quad a, q \ge 0.$$

The price of a is 1; the lesson price is $r > 0$; wealth is $m > 0$.

Must both goods be consumed? How could we reduce this to one variable?

Budget exhaustion gives $a = m - rq$ and $0 \le q \le m/r$.

Solve first, then respect the quantity bounds

$$\max_{0 \le q \le m/r} \; m - rq + 4\log(1+q).$$

Solve first, then respect the quantity bounds

$$\max_{0 \le q \le m/r} \; m - rq + 4\log(1+q).$$

The derivative $-r + 4/(1+q)$ is strictly decreasing.

Solve first, then respect the quantity bounds

NES

$$\max_{0 \le q \le m/r} \; m - rq + 4\log(1+q).$$

The derivative $-r + 4/(1+q)$ is strictly decreasing.

The stationary quantity is $4/r - 1$. Impose both bounds:

Solve first, then respect the quantity bounds

$$\max_{0 \le q \le m/r} \; m - rq + 4\log(1+q).$$

The derivative $-r + 4/(1+q)$ is strictly decreasing.

The stationary quantity is $4/r - 1$. Impose both bounds:

$$q^* = \min\left\{\frac{m}{r}, \max\left\{\frac{4}{r} - 1, 0\right\}\right\}, \quad a^* = m - rq^*.$$

Three regimes, one decision problem

If $r \geq 4$: no tutoring, $q^* = 0$, $a^* = m$.

Three regimes, one decision problem

If $r \geq 4$: no tutoring, $q^* = 0$, $a^* = m$.

If $0 < r < 4$ and $m \leq 4 - r$: the no-borrowing bound binds, $q^* = m/r$, $a^* = 0$.

Three regimes, one decision problem

If $r \geq 4$: no tutoring, $q^* = 0$, $a^* = m$.

If $0 < r < 4$ and $m \leq 4 - r$: the no-borrowing bound binds, $q^* = m/r$, $a^* = 0$.

If $0 < r < 4$ and $m > 4 - r$: both goods are consumed, $q^* = 4/r - 1$, $a^* = m - 4 + r$.

Three regimes, one decision problem

If $r \geq 4$: no tutoring, $q^* = 0$, $a^* = m$.

If $0 < r < 4$ and $m \leq 4 - r$: the no-borrowing bound binds, $q^* = m/r$, $a^* = 0$.

If $0 < r < 4$ and $m > 4 - r$: both goods are consumed, $q^* = 4/r - 1$, $a^* = m - 4 + r$.

At $r = 1$: income 2 gives $(a, q) = (0, 2)$; income 8 gives $(5, 3)$.

Three regimes, one decision problem

If $r \geq 4$: no tutoring, $q^* = 0$, $a^* = m$.

If $0 < r < 4$ and $m \leq 4 - r$: the no-borrowing bound binds, $q^* = m/r$, $a^* = 0$.

If $0 < r < 4$ and $m > 4 - r$: both goods are consumed, $q^* = 4/r - 1$, $a^* = m - 4 + r$.

At $r = 1$: income 2 gives $(a, q) = (0, 2)$; income 8 gives $(5, 3)$.

At $m = 4 - r$, both formulas agree; a binding bound may have zero multiplier.

Perfect complements: solve at the kink

$$u(x) = \min\{x_1, x_2\}.$$

Perfect complements: solve at the kink

$$u(x) = \min\{x_1, x_2\}.$$

What proportions would you buy, and why would a surplus be wasteful?

Perfect complements: solve at the kink

$$u(x) = \min\{x_1, x_2\}.$$

What proportions would you buy, and why would a surplus be wasteful?

Utility t costs at least $(p_1 + p_2)t$, with cheapest bundle (t, t).

Perfect complements: solve at the kink

$$u(x) = \min\{x_1, x_2\}.$$

What proportions would you buy, and why would a surplus be wasteful?

Utility t costs at least $(p_1 + p_2)t$, with cheapest bundle (t, t).

$$x_1^* = x_2^* = \frac{m}{p_1 + p_2}; \qquad p = (2, 1),\ m = 12 \ \Rightarrow\ (4, 4).$$

Perfect complements: solve at the kink

$$u(x) = \min\{x_1, x_2\}.$$

What proportions would you buy, and why would a surplus be wasteful?

Utility t costs at least $(p_1 + p_2)t$, with cheapest bundle (t, t).

$$x_1^* = x_2^* = \frac{m}{p_1 + p_2}; \qquad p = (2, 1),\ m = 12 \ \Rightarrow\ (4, 4).$$

The target-cost argument works even though smooth tangency fails at the kink.

Proof: convex preferences give a convex demand set

Assume preferences are convex and take $x, y \in D(p, m)$.

Proof: convex preferences give a convex demand set

NES

Assume preferences are convex and take $x, y \in D(p, m)$.

For $t \in [0, 1]$, $z_t = tx + (1 - t)y$ is affordable because $B(p, m)$ is convex.

Proof: convex preferences give a convex demand set

NES

Assume preferences are convex and take $x, y \in D(p, m)$.

For $t \in [0, 1]$, $z_t = tx + (1 - t)y$ is affordable because $B(p, m)$ is convex.

For every affordable w, optimality gives $x \succeq w$ and $y \succeq w$.

Proof: convex preferences give a convex demand set

NES

Assume preferences are convex and take $x, y \in D(p, m)$.

For $t \in [0, 1]$, $z_t = tx + (1 - t)y$ is affordable because $B(p, m)$ is convex.

For every affordable w, optimality gives $x \succeq w$ and $y \succeq w$.

Convex preferences imply $z_t \succeq w$ for every such w. Thus $z_t \in D(p, m)$.

Proof: convex preferences give a convex demand set

Assume preferences are convex and take $x, y \in D(p, m)$.

For $t \in [0, 1]$, $z_t = tx + (1 - t)y$ is affordable because $B(p, m)$ is convex.

For every affordable w, optimality gives $x \succeq w$ and $y \succeq w$.

Convex preferences imply $z_t \succeq w$ for every such w. Thus $z_t \in D(p, m)$.

Hence $D(p, m)$ is convex. It can still be a whole line segment.

Strictly convex preferences: uniqueness

Claim. On a convex budget set, strictly convex preferences give at most one optimum.

Strictly convex preferences: uniqueness

Claim. On a convex budget set, strictly convex preferences give at most one optimum.

Proof. If distinct x, y were optimal, they would be indifferent.

Strictly convex preferences: uniqueness

Claim. On a convex budget set, strictly convex preferences give at most one optimum.

Proof. If distinct x, y were optimal, they would be indifferent.

Their midpoint is affordable and, by strict convexity, strictly preferred: a contradiction.

Strictly convex preferences: uniqueness

Claim. On a convex budget set, strictly convex preferences give at most one optimum.

Proof. If distinct x, y were optimal, they would be indifferent.

Their midpoint is affordable and, by strict convexity, strictly preferred: a contradiction.

With existence already established, there is a unique demanded bundle.

Indirect utility: the optimal value

$$v(p, m) = \max_{x \geq 0,\, p \cdot x \leq m} u(x).$$

Indirect utility: the optimal value

$$v(p, m) = \max_{x \geq 0,\, p \cdot x \leq m} u(x).$$

Demand tells us **which bundle**; indirect utility tells us **the attained value**.

Indirect utility: the optimal value

$$v(p, m) = \max_{x \geq 0,\, p \cdot x \leq m} u(x).$$

Demand tells us **which bundle**; indirect utility tells us **the attained value**.

Even if $D(p, m)$ contains several bundles, every one gives the same value $v(p, m)$.

Indirect utility: the optimal value

$$v(p, m) = \max_{x \geq 0,\, p \cdot x \leq m} u(x).$$

Demand tells us **which bundle**; indirect utility tells us **the attained value**.

Even if $D(p, m)$ contains several bundles, every one gives the same value $v(p, m)$.

It is a function of the environment, not of a chosen bundle.

Compute the value: Cobb–Douglas

For $\widetilde{u} = \alpha \log x_1 + (1-\alpha) \log x_2$, substitute the optimal quantities:

Compute the value: Cobb–Douglas

For $\widetilde{u} = \alpha \log x_1 + (1-\alpha) \log x_2$, substitute the optimal quantities:

$$v_{\log} = \alpha \log \frac{\alpha m}{p_1} + (1-\alpha) \log \frac{(1-\alpha) m}{p_2}.$$

Compute the value: Cobb–Douglas

For $\widetilde{u} = \alpha \log x_1 + (1-\alpha) \log x_2$, substitute the optimal quantities:

$$v_{\log} = \alpha \log \frac{\alpha m}{p_1} + (1-\alpha) \log \frac{(1-\alpha) m}{p_2}.$$

Thus $v_{\log} = \log m - \alpha \log p_1 - (1-\alpha) \log p_2 + c_\alpha$, where $c_\alpha = \alpha \log \alpha + (1-\alpha) \log(1-\alpha)$.

Compute the value: Cobb–Douglas

For $\widetilde{u} = \alpha \log x_1 + (1-\alpha) \log x_2$, substitute the optimal quantities:

$$v_{\log} = \alpha \log \frac{\alpha m}{p_1} + (1-\alpha) \log \frac{(1-\alpha) m}{p_2}.$$

Thus $v_{\log} = \log m - \alpha \log p_1 - (1-\alpha) \log p_2 + c_\alpha$, where $c_\alpha = \alpha \log \alpha + (1-\alpha) \log(1-\alpha)$.

For the original product index, $v_{\text{raw}} = \exp(v_{\log})$. State which scale you use.

Compute the value: substitutes and complements

For $u = ax_1 + bx_2$, spend on the higher-return good:

Compute the value: substitutes and complements

For $u = ax_1 + bx_2$, spend on the higher-return good:

$$v(p, m) = m \max\left\{\frac{a}{p_1}, \frac{b}{p_2}\right\}.$$

Compute the value: substitutes and complements

NES

For $u = ax_1 + bx_2$, spend on the higher-return good:

$$v(p, m) = m \max\left\{\frac{a}{p_1}, \frac{b}{p_2}\right\}.$$

For $u = \min\{x_1, x_2\}$, use the equal-quantity optimum:

Compute the value: substitutes and complements

For $u = ax_1 + bx_2$, spend on the higher-return good:

$$v(p, m) = m \max\left\{\frac{a}{p_1}, \frac{b}{p_2}\right\}.$$

For $u = \min\{x_1, x_2\}$, use the equal-quantity optimum:

$$v(p, m) = \frac{m}{p_1 + p_2}.$$

Compute the value: the logarithmic model

Return to $u = a + 4\log(1+q)$ and $a + rq \le m$. Substitute the demands:

Compute the value: the logarithmic model

Return to $u = a + 4\log(1+q)$ and $a + rq \le m$. Substitute the demands:

$$v(r,m) = \begin{cases} m, & r \ge 4, \\ 4\log(1+m/r), & 0 < r < 4,\ m \le 4 - r, \\ m - 4 + r + 4\log(4/r), & 0 < r < 4,\ m > 4 - r. \end{cases}$$

Compute the value: the logarithmic model

Return to $u = a + 4\log(1+q)$ and $a + rq \le m$. Substitute the demands:

$$v(r,m) = \begin{cases} m, & r \ge 4, \\ 4\log(1+m/r), & 0 < r < 4,\ m \le 4 - r, \\ m - 4 + r + 4\log(4/r), & 0 < r < 4,\ m > 4 - r. \end{cases}$$

What happens to this value when wealth increases or the lesson price rises?

Indirect utility: the properties to prove

- Maintain continuous u, $p \gg 0$ and $m > 0$, so optimal values exist.

Indirect utility: the properties to prove

NES

- Maintain continuous u, $p \gg 0$ and $m > 0$, so optimal values exist.
- $v(tp, tm) = v(p, m)$; higher prices cannot increase v.

Indirect utility: the properties to prove

NES

- Maintain continuous u, $p \gg 0$ and $m > 0$, so optimal values exist.
- $v(tp, tm) = v(p, m)$; higher prices cannot increase v.
- More wealth weakly increases v; under LNS the increase is strict.

Indirect utility: the properties to prove

NES

- Maintain continuous u, $p \gg 0$ and $m > 0$, so optimal values exist.
- $v(tp, tm) = v(p, m)$; higher prices cannot increase v.
- More wealth weakly increases v; under LNS the increase is strict.
- v is continuous on the positive-price, positive-wealth domain.

A less obvious property: quasiconvexity

NES

For $0 \leq t \leq 1$, set $(p^t, m^t) = t(p, m) + (1 - t)(q, n)$.

A less obvious property: quasiconvexity

NES

For $0 \leq t \leq 1$, set $(p^t, m^t) = t(p, m) + (1 - t)(q, n)$.

$$v(p^t, m^t) \leq \max\{v(p, m), v(q, n)\}.$$

A less obvious property: quasiconvexity

NES

For $0 \leq t \leq 1$, set $(p^t, m^t) = t(p, m) + (1 - t)(q, n)$.

$$v(p^t, m^t) \leq \max\{v(p, m), v(q, n)\}.$$

If a bundle is unaffordable at both endpoints, it is unaffordable at their mixture too.

A less obvious property: quasiconvexity

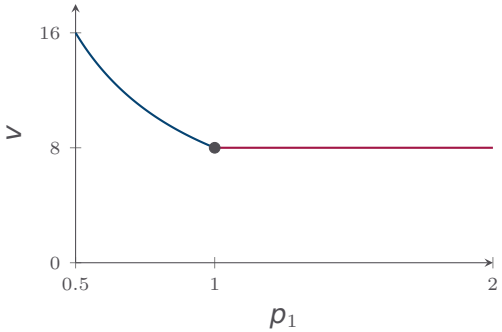
For $0 \leq t \leq 1$, set $(p^t, m^t) = t(p, m) + (1 - t)(q, n)$.

$$v(p^t, m^t) \leq \max\{v(p, m), v(q, n)\}.$$

If a bundle is unaffordable at both endpoints, it is unaffordable at their mixture too.

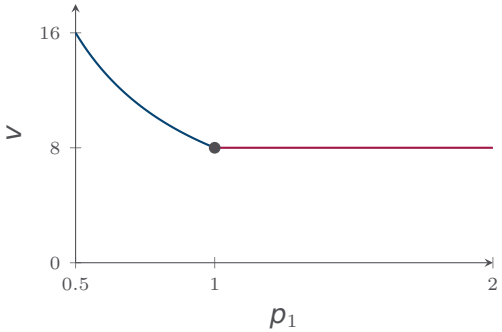
So the mixed budget cannot deliver utility above both endpoint values.

A continuous value need not be differentiable



$u = x_1 + x_2$, $p_2 = 1$, $m = 8$.

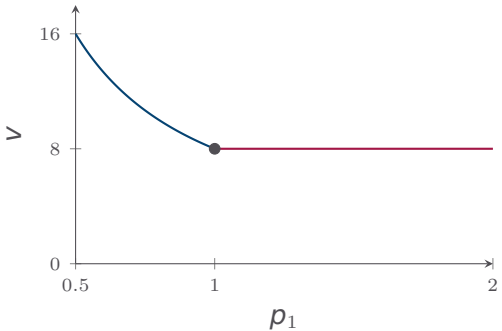
A continuous value need not be differentiable



$u = x_1 + x_2$, $p_2 = 1$, $m = 8$.

$$v(p_1, 1, 8) = \frac{8}{\min\{p_1, 1\}}.$$

A continuous value need not be differentiable



$u = x_1 + x_2$, $p_2 = 1$, $m = 8$.

$$v(p_1, 1, 8) = \frac{8}{\min\{p_1, 1\}}.$$

At $p_1 = 1$, the consumer is indifferent along the budget line. The value has a kink.

A general parameterised optimisation problem

$$V(\theta) = \max_x f(x, \theta) \quad \text{subject to} \quad g_j(x, \theta) \geq 0, \quad h_k(x, \theta) = 0.$$

A general parameterised optimisation problem

$$V(\theta) = \max_x f(x, \theta) \quad \text{subject to} \quad g_j(x, \theta) \geq 0, \quad h_k(x, \theta) = 0.$$

The parameter varies in an open set; objective and constraints are continuously differentiable.

A general parameterised optimisation problem

$$V(\theta) = \max_{x} f(x, \theta) \quad \text{subject to} \quad g_j(x, \theta) \geq 0, \quad h_k(x, \theta) = 0.$$

The parameter varies in an open set; objective and constraints are continuously differentiable.

The ambient decision domain is open; all quantity bounds are included among the constraints.

A general parameterised optimisation problem

$$V(\theta) = \max_x f(x, \theta) \quad \text{subject to} \quad g_j(x, \theta) \geq 0, \quad h_k(x, \theta) = 0.$$

The parameter varies in an open set; objective and constraints are continuously differentiable.

The ambient decision domain is open; all quantity bounds are included among the constraints.

Use $\lambda_j \geq 0$ for inequalities and unrestricted ν_k for equalities.

The general Lagrangian

$$\mathcal{L} = f(x, \theta) + \sum_j \lambda_j g_j(x, \theta) + \sum_k \nu_k h_k(x, \theta).$$

The general Lagrangian

$$\mathcal{L} = f(x, \theta) + \sum_j \lambda_j g_j(x, \theta) + \sum_k \nu_k h_k(x, \theta).$$

At an optimum: $\mathcal{L}_x = 0$, $g_j \geq 0$, $h_k = 0$, $\lambda_j \geq 0$ and $\lambda_j g_j = 0$.

The general Lagrangian

$$\mathcal{L} = f(x, \theta) + \sum_j \lambda_j g_j(x, \theta) + \sum_k \nu_k h_k(x, \theta).$$

At an optimum: $\mathcal{L}_x = 0$, $g_j \geq 0$, $h_k = 0$, $\lambda_j \geq 0$ and $\lambda_j g_j = 0$.

Assume a continuously differentiable selection of optimal choices and KKT multipliers exists near the parameter under consideration.

The general Lagrangian

$$\mathcal{L} = f(x, \theta) + \sum_j \lambda_j g_j(x, \theta) + \sum_k \nu_k h_k(x, \theta).$$

At an optimum: $\mathcal{L}_x = 0$, $g_j \geq 0$, $h_k = 0$, $\lambda_j \geq 0$ and $\lambda_j g_j = 0$.

Assume a continuously differentiable selection of optimal choices and KKT multipliers exists near the parameter under consideration.

This is a sufficient smooth-branch regularity assumption, to be checked in an application.

Envelope theorem: a constrained smooth version

Under the preceding smooth-branch assumptions,

Envelope theorem: a constrained smooth version

Under the preceding smooth-branch assumptions,

$$\frac{\partial V}{\partial \theta_\ell} = \frac{\partial \mathcal{L}}{\partial \theta_\ell}(x^*, \lambda^*, \nu^*; \theta).$$

Envelope theorem: a constrained smooth version

Under the preceding smooth-branch assumptions,

$$\frac{\partial V}{\partial \theta_\ell} = \frac{\partial \mathcal{L}}{\partial \theta_\ell}(x^*, \lambda^*, \nu^*; \theta).$$

On the right, hold decisions and multipliers fixed when taking the partial derivative.

Envelope theorem: a constrained smooth version

Under the preceding smooth-branch assumptions,

$$\frac{\partial V}{\partial \theta_\ell} = \frac{\partial \mathcal{L}}{\partial \theta_\ell}(x^*, \lambda^*, \nu^*; \theta).$$

On the right, hold decisions and multipliers fixed when taking the partial derivative.

Thus $V_{\theta_\ell} = f_{\theta_\ell} + \sum_j \lambda_j^* (g_j)_{\theta_\ell} + \sum_k \nu_k^* (h_k)_{\theta_\ell}$, evaluated at the optimum.

Proof: differentiate along an optimal branch

Feasibility and slackness give $V(\theta) = \mathcal{L}(x^*(\theta), \lambda^*(\theta), \nu^*(\theta); \theta)$.

Proof: differentiate along an optimal branch

Feasibility and slackness give $V(\theta) = \mathcal{L}(x^*(\theta), \lambda^*(\theta), \nu^*(\theta); \theta)$.

Fix one coordinate t of θ. The chain rule gives

Proof: differentiate along an optimal branch

Feasibility and slackness give $V(\theta) = \mathcal{L}(x^*(\theta), \lambda^*(\theta), \nu^*(\theta); \theta)$.

Fix one coordinate t of θ. The chain rule gives

$$\frac{dV}{dt} = \mathcal{L}_t + \mathcal{L}_x x_t^* + \sum_j g_j(\lambda_j^*)_t + \sum_k h_k(\nu_k^*)_t.$$

Proof: differentiate along an optimal branch

Feasibility and slackness give $V(\theta) = \mathcal{L}(x^*(\theta), \lambda^*(\theta), \nu^*(\theta); \theta)$.

Fix one coordinate t of θ. The chain rule gives

$$\frac{dV}{dt} = \mathcal{L}_t + \mathcal{L}_x x_t^* + \sum_j g_j(\lambda_j^*)_t + \sum_k h_k(\nu_k^*)_t.$$

Stationarity kills the $\mathcal{L}_x$ term; equality feasibility kills the h_k terms.

Proof: why multiplier derivatives disappear

For each inequality, examine $g_j(\lambda_j^*)_t$.

Proof: why multiplier derivatives disappear

For each inequality, examine $g_j(\lambda_j^*)_t$.

If $\lambda_j^* > 0$, complementary slackness gives $g_j = 0$.

Proof: why multiplier derivatives disappear

For each inequality, examine $g_j(\lambda_j^*)_t$.

If $\lambda_j^* > 0$, complementary slackness gives $g_j = 0$.

If $\lambda_j^* = 0$, its differentiable nonnegative branch has a local minimum at zero, so $(\lambda_j^*)_t = 0$.

Proof: why multiplier derivatives disappear

NES

For each inequality, examine $g_j(\lambda_j^*)_t$.

If $\lambda_j^* > 0$, complementary slackness gives $g_j = 0$.

If $\lambda_j^* = 0$, its differentiable nonnegative branch has a local minimum at zero, so $(\lambda_j^*)_t = 0$.

Every extra term is zero. Hence

$$\frac{dV}{dt} = \mathcal{L}_t.$$

Income and price derivatives of indirect utility

The parameter-dependent constraint is $m - p \cdot x \geq 0$. Quantity bounds do not depend on prices or wealth.

Income and price derivatives of indirect utility

The parameter-dependent constraint is $m - p \cdot x \geq 0$. Quantity bounds do not depend on prices or wealth.

$$v_m = \mathcal{L}_m = \lambda^*.$$

Income and price derivatives of indirect utility

The parameter-dependent constraint is $m - p \cdot x \geq 0$. Quantity bounds do not depend on prices or wealth.

$$v_m = \mathcal{L}_m = \lambda^*.$$

$$v_{p_i} = \mathcal{L}_{p_i} = -\lambda^* x_i^*.$$

Income and price derivatives of indirect utility

The parameter-dependent constraint is $m - p \cdot x \geq 0$. Quantity bounds do not depend on prices or wealth.

$$v_m = \mathcal{L}_m = \lambda^*.$$

$$v_{p_i} = \mathcal{L}_{p_i} = -\lambda^* x_i^*.$$

Where the envelope assumptions hold, λ^* is the marginal value of wealth in the chosen utility units.

Roy’s identity: recover demand from value

Assume continuous utility, LNS, $p \gg 0$, $m > 0$, differentiable v and $v_m > 0$.

Roy's identity: recover demand from value

Assume continuous utility, LNS, $p \gg 0$, $m > 0$, differentiable v and $v_m > 0$.

Every optimiser satisfies

$$x_i^* = -\frac{v_{p_i}(p, m)}{v_m(p, m)}.$$

Roy's identity: recover demand from value

Assume continuous utility, LNS, $p \gg 0$, $m > 0$, differentiable v and $v_m > 0$.

Every optimiser satisfies

$$x_i^* = -\frac{v_{p_i}(p, m)}{v_m(p, m)}.$$

At a smooth regular optimum, the envelope formulas give the ratio by cancelling λ^*.

Roy's identity: recover demand from value

Assume continuous utility, LNS, $p \gg 0$, $m > 0$, differentiable v and $v_m > 0$.

Every optimiser satisfies

$$x_i^* = -\frac{v_{p_i}(p, m)}{v_m(p, m)}.$$

At a smooth regular optimum, the envelope formulas give the ratio by cancelling λ^*.

The next proof needs no differentiable demand or differentiable direct utility.

Roy without differentiating demand

Let $x^* \in D(p, m)$. LNS gives $p \cdot x^* = m$.

→ Back

Roy without differentiating demand

Let $x^* \in D(p, m)$. LNS gives $p \cdot x^* = m$.

At any nearby price vector q, the bundle x^* is affordable with wealth $q \cdot x^*$. Thus $F(q) = v(q, q \cdot x^*) \geq u(x^*) = F(p)$.

→ Back

Roy without differentiating demand

Let $x^* \in D(p, m)$. LNS gives $p \cdot x^* = m$.

At any nearby price vector q, the bundle x^* is affordable with wealth $q \cdot x^*$. Thus $F(q) = v(q, q \cdot x^*) \geq u(x^*) = F(p)$.

If v is differentiable, F has a minimum at p:

$$0 = F_{q_i}(p) = v_{p_i}(p, m) + v_m(p, m) x_i^*.$$

→ Back

Roy without differentiating demand

Let $x^* \in D(p, m)$. LNS gives $p \cdot x^* = m$.

At any nearby price vector q, the bundle x^* is affordable with wealth $q \cdot x^*$. Thus $F(q) = v(q, q \cdot x^*) \geq u(x^*) = F(p)$.

If v is differentiable, F has a minimum at p:

$$0 = F_{q_i}(p) = v_{p_i}(p, m) + v_m(p, m) x_i^*.$$

If $v_m > 0$, every optimiser satisfies the same Roy formula, so demand is unique.

→ Back

Check Roy in the Cobb–Douglas example

$$v = \log m - \alpha \log p_1 - (1-\alpha) \log p_2 + c_\alpha.$$

Check Roy in the Cobb–Douglas example

NES

$$v = \log m - \alpha \log p_1 - (1-\alpha) \log p_2 + c_\alpha.$$

$$v_m = 1/m, \qquad v_{p_1} = -\alpha/p_1, \qquad v_{p_2} = -(1-\alpha)/p_2.$$

Check Roy in the Cobb–Douglas example

$$v = \log m - \alpha \log p_1 - (1-\alpha)\log p_2 + c_\alpha.$$

$$v_m = 1/m, \qquad v_{p_1} = -\alpha/p_1, \qquad v_{p_2} = -(1-\alpha)/p_2.$$

$$x_1 = -\frac{-\alpha/p_1}{1/m} = \frac{\alpha m}{p_1}.$$

Check Roy in the Cobb–Douglas example

NES

$$v = \log m - \alpha \log p_1 - (1-\alpha)\log p_2 + c_\alpha.$$

$$v_m = 1/m, \qquad v_{p_1} = -\alpha/p_1, \qquad v_{p_2} = -(1-\alpha)/p_2.$$

$$x_1 = -\frac{-\alpha/p_1}{1/m} = \frac{\alpha m}{p_1}.$$

Likewise $x_2 = (1-\alpha)m/p_2$. The demand agrees with the direct solution.

Roy example 2: substitutes and a corner

NES

For $u = ax_1 + bx_2$,

$$v(p, m) = m \max\left\{\frac{a}{p_1}, \frac{b}{p_2}\right\}.$$

Roy example 2: substitutes and a corner

For $u = ax_1 + bx_2$,

$$v(p, m) = m \max\left\{\frac{a}{p_1}, \frac{b}{p_2}\right\}.$$

Differentiate in a region where good 1 gives the strictly larger spending return.

Roy example 2: substitutes and a corner

For $u = ax_1 + bx_2$,

$$v(p, m) = m \max\left\{\frac{a}{p_1}, \frac{b}{p_2}\right\}.$$

Differentiate in a region where good 1 gives the strictly larger spending return.

If $a/p_1 > b/p_2$, then $v = am/p_1$, $v_m = a/p_1$, $v_{p_1} = -am/p_1^2$ and $v_{p_2} = 0$.

Roy example 2: substitutes and a corner

For $u = ax_1 + bx_2$,

$$v(p, m) = m \max\left\{\frac{a}{p_1}, \frac{b}{p_2}\right\}.$$

Differentiate in a region where good 1 gives the strictly larger spending return.

If $a/p_1 > b/p_2$, then $v = am/p_1$, $v_m = a/p_1$, $v_{p_1} = -am/p_1^2$ and $v_{p_2} = 0$.

Roy gives $x_1 = m/p_1$, $x_2 = 0$.

Roy example 4: a kink in direct utility

NES

For $u = \min\{x_1, x_2\}$,

$$v(p, m) = \frac{m}{p_1 + p_2}.$$

Roy example 4: a kink in direct utility

For $u = \min\{x_1, x_2\}$,

$$v(p, m) = \frac{m}{p_1 + p_2}.$$

$v_m = 1/(p_1 + p_2)$ and $v_{p_i} = -m/(p_1 + p_2)^2$.

Roy example 4: a kink in direct utility

For $u = \min\{x_1, x_2\}$,

$$v(p, m) = \frac{m}{p_1 + p_2}.$$

$v_m = 1/(p_1 + p_2)$ and $v_{p_i} = -m/(p_1 + p_2)^2$.

$$x_i = -\frac{v_{p_i}}{v_m} = \frac{m}{p_1 + p_2} \qquad (i = 1, 2).$$

Roy example 4: a kink in direct utility

For $u = \min\{x_1, x_2\}$,

$$v(p, m) = \frac{m}{p_1 + p_2}.$$

$v_m = 1/(p_1 + p_2)$ and $v_{p_i} = -m/(p_1 + p_2)^2$.

$$x_i = -\frac{v_{p_i}}{v_m} = \frac{m}{p_1 + p_2} \qquad (i = 1, 2).$$

Direct utility has a kink, but indirect utility is differentiable. The direct proof of Roy still applies.

One ratio, three regimes

For the logarithmic value computed earlier, use $q^* = -v_r / v_m$.

Regime	v_m	v_r	$-v_r / v_m$

One ratio, three regimes

For the logarithmic value computed earlier, use $q^* = -v_r/v_m$.

Regime	v_m	v_r	$-v_r/v_m$
No tutoring	1	0	0

One ratio, three regimes

For the logarithmic value computed earlier, use $q^* = -v_r/v_m$.

Regime	v_m	v_r	$-v_r/v_m$
No tutoring	1	0	0
All wealth on q	$\frac{4}{r+m}$	$-\frac{4m}{r(r+m)}$	$\frac{m}{r}$

One ratio, three regimes

For the logarithmic value computed earlier, use $q^* = -v_r/v_m$.

Regime	v_m	v_r	$-v_r/v_m$
No tutoring	1	0	0
All wealth on q	$\frac{4}{r+m}$	$-\frac{4m}{r(r+m)}$	$\frac{m}{r}$
Both goods	1	$1-\frac{4}{r}$	$\frac{4}{r}-1$

One ratio, three regimes

NES

For the logarithmic value computed earlier, use $q^* = -v_r/v_m$.

Regime	v_m	v_r	$-v_r/v_m$
No tutoring	1	0	0
All wealth on q	$\frac{4}{r+m}$	$-\frac{4m}{r(r+m)}$	$\frac{m}{r}$
Both goods	1	$1-\frac{4}{r}$	$\frac{4}{r}-1$

Recover $a^* = m - rq^*$. At $m = 4 - r$ and at $r = 4$, adjacent values and first derivatives agree.

Keep the tools, and read next

Varian, 9th edition: Chapter 5 *Choice*, Chapter 6 *Demand*.

Keep the tools, and read next

Varian, 9th edition: Chapter 5 *Choice*, Chapter 6 *Demand*.

Optional: Mas-Colell, Whinston and Green, section 3.D and the Mathematical Appendix on constrained optimisation.

Keep the tools, and read next

Varian, 9th edition: Chapter 5 *Choice*, Chapter 6 *Demand*.

Optional: Mas-Colell, Whinston and Green, section 3.D and the Mathematical Appendix on constrained optimisation.

Next lecture: expenditure minimisation, Hicksian demand and duality. Homework 1 is due 14 November 2026.

Next block: two ways to ask the consumer question

Utility maximisation: given p, m, how much utility can the consumer attain?

Next block: two ways to ask the consumer question

Utility maximisation: given p, m, how much utility can the consumer attain?

Expenditure minimisation: given $p, \bar{u}$, how little spending attains the target?

Next block: two ways to ask the consumer question

Utility maximisation: given p, m, how much utility can the consumer attain?

Expenditure minimisation: given $p, \bar{u}$, how little spending attains the target?

One fixes resources; the other fixes a level of the same preferences.

Expenditure minimisation

$$e(p, \bar{u}) = \min_{x \geq 0,\, u(x) \geq \bar{u}} p \cdot x.$$

Expenditure minimisation

$$e(p, \bar{u}) = \min_{x \geq 0,\, u(x) \geq \bar{u}} p \cdot x.$$

$$H(p, \bar{u}) = \arg\min_{x \geq 0,\, u(x) \geq \bar{u}} p \cdot x.$$

Expenditure minimisation

$$e(p, \bar{u}) = \min_{x \geq 0,\, u(x) \geq \bar{u}} p \cdot x.$$

$$H(p, \bar{u}) = \arg\min_{x \geq 0,\, u(x) \geq \bar{u}} p \cdot x.$$

e is the expenditure function; H is the Hicksian demand correspondence.

Expenditure minimisation

$$e(p, \bar{u}) = \min_{x \geq 0,\, u(x) \geq \bar{u}} p \cdot x.$$

$$H(p, \bar{u}) = \arg\min_{x \geq 0,\, u(x) \geq \bar{u}} p \cdot x.$$

e is the expenditure function; H is the Hicksian demand correspondence.

Assume the target is attainable. A singleton is denoted $h(p, \bar{u})$.

Why an unbounded feasible set can still have a minimum

Assume $p \gg 0$, continuous u, and pick a bundle $\widehat{x}$ with $u(\widehat{x}) \geq \bar{u}$.

Why an unbounded feasible set can still have a minimum

Assume $p \gg 0$, continuous u, and pick a bundle $\widehat{x}$ with $u(\widehat{x}) \geq \bar{u}$.

Only consider target-feasible bundles with $p \cdot x \leq p \cdot \widehat{x}$.

Why an unbounded feasible set can still have a minimum

Assume $p \gg 0$, continuous u, and pick a bundle $\widehat{x}$ with $u(\widehat{x}) \geq \bar{u}$.

Only consider target-feasible bundles with $p \cdot x \leq p \cdot \widehat{x}$.

This set is nonempty, closed and bounded, hence compact.

Why an unbounded feasible set can still have a minimum

Assume $p \gg 0$, continuous u, and pick a bundle $\widehat{x}$ with $u(\widehat{x}) \geq \bar{u}$.

Only consider target-feasible bundles with $p \cdot x \leq p \cdot \widehat{x}$.

This set is nonempty, closed and bounded, hence compact.

A continuous cost attains a minimum there; excluded bundles are more expensive.

Expenditure: a few properties for next time

NES

$e(tp, \bar{u}) = te(p, \bar{u})$ for $t > 0$; raising prices componentwise cannot lower expenditure.

Expenditure: a few properties for next time

NES

$e(tp, \bar{u}) = te(p, \bar{u})$ for $t > 0$; raising prices componentwise cannot lower expenditure.

For a fixed attainable target, expenditure is concave in prices:

Expenditure: a few properties for next time

NES

$e(tp, \bar{u}) = te(p, \bar{u})$ for $t > 0$; raising prices componentwise cannot lower expenditure.

For a fixed attainable target, expenditure is concave in prices:

$$e(tp + (1-t)q, \bar{u}) \geq te(p, \bar{u}) + (1-t)e(q, \bar{u}).$$

Expenditure: a few properties for next time

$e(tp, \bar{u}) = te(p, \bar{u})$ for $t > 0$; raising prices componentwise cannot lower expenditure.

For a fixed attainable target, expenditure is concave in prices:

$$e(tp + (1-t)q, \bar{u}) \geq te(p, \bar{u}) + (1-t)e(q, \bar{u}).$$

It is the minimum over target-feasible bundles of linear price functions.

Two identities to develop next time

Under continuous LNS preferences, positive prices and an attainable target $\bar{u} > u(0)$:

Two identities to develop next time

Under continuous LNS preferences, positive prices and an attainable target $\bar{u} > u(0)$:

$$e\big(p, v(p,m)\big) = m, \qquad v\big(p, e(p,\bar{u})\big) = \bar{u}.$$

Two identities to develop next time

NES

Under continuous LNS preferences, positive prices and an attainable target $\bar{u} > u(0)$:

$$e\big(p, v(p,m)\big) = m, \qquad v\big(p, e(p,\bar{u})\big) = \bar{u}.$$

For unique demands,

$$h\big(p, v(p,m)\big) = x(p,m).$$

Two identities to develop next time

Under continuous LNS preferences, positive prices and an attainable target $\bar{u} > u(0)$:

$$e\big(p, v(p,m)\big) = m, \qquad v\big(p, e(p,\bar{u})\big) = \bar{u}.$$

For unique demands,

$$h\big(p, v(p,m)\big) = x(p,m).$$

These connect the two problems; they are not an assumption that the consumer literally solves both.